

[8] Connectors

DESIGN NOTE:  
J1 host header (JST-XH, 7 pins):  
1 +3.3V 2 CS 3 DIN 4 DOUT/RDY  
5 SYNC 6 SCLK 7 GND

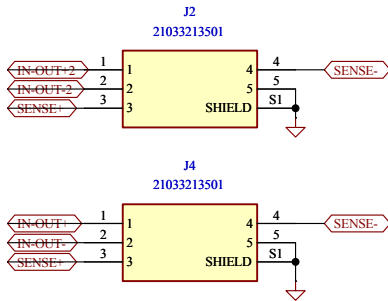
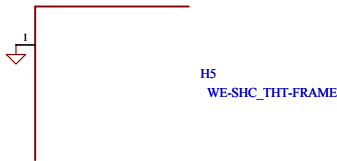
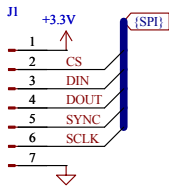
DESIGN NOTE:  
J2/J4 load cell (5 pins):  
1 OUT, 2 OUT (signal pair), 3 EXC+ (SENSE+),  
4 EXC- (SENSE-/BPDSW), 5 GND/shield.  
Both channels read V(pin2) - V(pin1):  
fix the sign in firmware if needed.

LAYOUT NOTE:  
H5 shielding frame covers the AD7190  
front-end; connector shells tied to GND.

Pinout and notes



Power flags: +3.3V and GND come from the host connector J1,  
(J2/J4 21033213501, H5 WE-SHC).



Host and load-cell connectors



|                                                                                                 |                                           |                                     |                                     |                                    |
|-------------------------------------------------------------------------------------------------|-------------------------------------------|-------------------------------------|-------------------------------------|------------------------------------|
| <b>Comments:</b><br><b>References:</b><br>JST XH B7B-XH-AM (J1)<br>J2/J4 21033213501, H5 WE-SHC | <b>Company:</b><br>OPEN THRUST LAB        |                                     | <b>Variant:</b>                     | <b>Git Hash:</b><br>b3867f9        |
|                                                                                                 | <b>Board Name:</b><br>AD7190              |                                     | <b>Project Name:</b><br>OPEN_WEIGHT |                                    |
| <b>Sheet Title:</b><br>Connectors                                                               | <b>File Name:</b><br>Connectors.kicad_sch | <b>Designer:</b><br>SIEBERT DIMITRY | <b>Date:</b><br>2026-09-29          | <b>Revision:</b><br>+ (Unreleased) |
| <b>Sheet Path:</b><br>/Project Architecture/Connectors/                                         |                                           | <b>Reviewer:</b>                    | <b>Size:</b><br>A3                  | <b>Sheet:</b><br>8 of 10           |